

LECTURE 7

Topics

Stock valuation models, the discounted dividend model (DDM), valuation analysis using multiples, the discounted cash flow model (DCF), fixed income security valuation

- The equity holders of a firm are the residual claimants on the assets of the firm
- The main form of income that common shareholders expect to receive from owning the common equity of a firm is *dividend distributions*
- The firm has complete discretion over the modality of the dividend distributions:
 - The timing of when the firm will initiate, suspend, or reinstate dividend distributions
 - The amount of the dividend distribution (regular or extraordinary)
- From the perspective of common shareholders, dividend distributions are *uncertain* or risky because they don't know the amount that will be distributed, and when the firm will suspend or reinstate the dividend distributions

- Suppose that a firm committed today ($t = 0$) to pay out a dividend, d_1 , at the end of the current period ($t = 1$)
- For the periods $t > 1$ onwards, shareholders do not know whether the firm will continue to make dividend distributions and the amount of these distributions
- We can model the dividends at period $t = 2$ as follows:

$$\tilde{d}_2 = d_1 \times (1 + \tilde{g}_1),$$

where $\tilde{g}_1$ is the *dividend growth rate*

- The dividend growth rate $\tilde{g}_1$ is a random variable with a known distribution, F_g , with a mean $\bar{g}$ and standard deviation σ_g
- In particular, $\tilde{g}_1 = -1$, if the firm decides to suspend the dividend distribution
- $\tilde{g}_1 = 0$, if the firm decides to leave unchanged the current dividend payout policy or initiate a new dividend distribution

- We can now extend this framework to characterize the dividend payment at some future date $t = N$ as follows:

$$\tilde{d}_N = d_1 \times (1 + \tilde{g}_1) \times (1 + \tilde{g}_2) \times \dots \times (1 + \tilde{g}_{N-1}),$$

- The dividend growth rates $\tilde{g}_2, \dots, \tilde{g}_{N-1}$ are random variables with same distribution, F_g , with mean $\bar{g}$ and standard deviation σ_g
- Thus, the stream of cash flows that common shareholders expect to receive for their investment is:

Period	0	1	2	...	N	...
CFs		d_1	$\tilde{d}_2$	...	$\tilde{d}_{N-1}$	...

- The value of the firm's equity is the present value of this stream of future dividends
- What discount rate should shareholders use?

- In corporate finance you learned that the discount rate that we use to calculate the present value of a stream of cash flows must reflect the *risk* of those cash flows
- In our case, the cash flows are the firm's dividends
- The risk of a firm's dividends mirror the risk of the firm's equity
- In finance, we can map *risk* into *rates* using asset pricing models
- For simplicity, let's use the CAPM
- Suppose the firm's equity beta is β
- Then the expected rate of return corresponding to this beta can be calculated using the CAPM formula:

$$k = r_F + \beta(E(r_M) - r_F)$$

- This rate is known as the shareholders' *required rate of return*

- We can now calculate the present value of the stream of dividends:

$$\hat{P}_0 = \frac{d_1}{1+k} + \frac{E(\tilde{d}_2)}{(1+k)^2} + \dots + \frac{E(\tilde{d}_N)}{(1+k)^N} + \dots$$

- The value $\hat{P}_0$ is known as the *intrinsic value* of the firm's equity
- If the actual value of the firm's equity at time $t = 0$ is P_0 , then we can compare this value against the intrinsic value:
 - If $\hat{P}_0 = P_0$, we say that the firm's stock is *fairly-valued*
 - If $\hat{P}_0 > P_0$, we say that the firm's stock is *undervalued*
 - If $\hat{P}_0 < P_0$, we say that the firm's stock is *overvalued*
- Investors want to *buy* undervalued stocks, *sell* overvalued stocks and *hold* fairly-valued stocks

- **Assumption:** Suppose now that the dividend growth rates $\tilde{g}_1, \tilde{g}_2, \dots, \tilde{g}_N, \dots$ are *independent* of each other (i.e., the correlation between any two such rates is zero)
- Then, the expected dividends become:

$$E(\tilde{d}_2) = E(d_1 \times (1 + \tilde{g}_1)) = d_1 \times (1 + \bar{g})$$

$$E(\tilde{d}_3) = E(d_1 \times (1 + \tilde{g}_1) \times (1 + \tilde{g}_2)) = d_1 \times (1 + \bar{g})^2$$

.....

$$E(\tilde{d}_N) = E(d_1 \times (1 + \tilde{g}_1) \times \dots \times (1 + \tilde{g}_{N-1})) = d_1 \times (1 + \bar{g})^{N-1}$$

.....

- The intrinsic value of the firm's equity becomes:

$$\begin{aligned}\hat{P}_0 &= \frac{d_1}{1+k} + \frac{E(\tilde{d}_2)}{(1+k)^2} + \dots + \frac{E(\tilde{d}_N)}{(1+k)^N} + \dots \\ &= \frac{d_1}{1+k} + \frac{d_1 \times (1+\bar{g})}{(1+k)^2} + \dots + \frac{d_1 \times (1+\bar{g})^{N-1}}{(1+k)^N} + \dots \\ &= \frac{d_1}{k - \bar{g}}\end{aligned}$$

- This model is the Gordon's discounted dividend model (DDM)

Gordon's discounted dividend model (DDM)

When the dividend growth satisfy the conditions under the assumptions above, the intrinsic value of a firm's stock becomes:

$$\hat{P}_0 = \frac{d_1}{k - \bar{g}},$$

where k is the required rate and $\bar{g}$ is the mean dividend growth rate

- We can use Gordon's DDM to infer the alpha of the stock
- Recall that a stock's alpha is the main input in the Treynor-Black model to find the optimal portfolio
- To find the stock's alpha implied by the DDM, we first set the intrinsic value of the stock to the stock price and solve for k

$$P_0 = \hat{P}_0 = \frac{d_1}{k - \bar{g}}$$

and solving for k we obtain:

$$k = \underbrace{\bar{g}}_{\text{dividend growth}} + \underbrace{\frac{d_1}{P_0}}_{\text{dividend yield}}$$

- If the stock's beta is β , then the alpha implied by the DDM is:

$$\alpha^{DDM} = \underbrace{\left(\bar{g} + \frac{d_1}{P_0} \right)}_k - \underbrace{(r_F + \beta(E(r_M) - r_F))}_{k^{CAPM}}$$

- α^{DDM} tells us by how much the stock market undervalues or overvalues the stock relative to its intrinsic value
 - If $\alpha^{DDM} > 0$, the stock is undervalued and investors should buy
 - If $\alpha^{DDM} < 0$, the stock is overvalued and investors should sell or short-sell (if they don't already own the stock)
 - If $\alpha^{DDM} = 0$, the stock is fairly-valued and investors should hold

- We can make Gordon's DDM a bit more general, if we take into account a firm's *business life-cycle*
- Most firms will go through 5 phases of development: Seed, Start-up, Growth, Maturity, Stability/Decline
- How does the DDM change to account for this life-cycle phases?
- Suppose a firm is currently in a Growth phase and is expected to make a series of *extraordinary* dividend payments until it reaches the *horizon* for the Maturity phase, $t = H$
- Suppose the average extraordinary dividends are: $\bar{d}_1, \bar{d}_2, \dots, \bar{d}_H$
- Starting with the horizon $t = H$, the dividends are expected to become more regular and grow at an average rate of $\bar{g}$
- Finally, suppose that the required rate of return is k

- Then, the intrinsic value of the firm's stock at $t = 0$ becomes:

$$\hat{P}_0 = \underbrace{\frac{\bar{d}_1}{1+k} + \dots + \frac{\bar{d}_H}{(1+k)^H}}_{\text{PV of extraordinary dividends}} + \underbrace{\frac{1}{(1+k)^H} \times \hat{P}_H}_{\text{PV of the stock's intrinsic value at the horizon}}$$

- The stock's intrinsic value at the horizon becomes can now be calculated using the original DDM formula
- First, note that the average dividend at $t = H + 1$ is $\bar{d}_{H+1} = \bar{d}_H \times (1 + \bar{g})$
- Then, the intrinsic value of the stock at the horizon is:

$$\hat{P}_H = \frac{\bar{d}_H \times (1 + \bar{g})}{k - \bar{g}}$$

- Dividends are the portion of the firm's *earning* (also known as *net income*) that the firm's management chooses to distribute back to the shareholders
- The portion of the earnings that is not distributed in dividends is called *retained earnings* and is the portion of the earnings that is *plowing* back (i.e., invests back) into the assets of the firm
- Thus, we can think of a firm's earnings as being decomposed into two components:

$$\underbrace{NI}_{\text{earnings or net income}} = \underbrace{d}_{\text{dividends}} + \underbrace{RE}_{\text{retained earnings}}$$

for any period t .

- *Plowback ratio*, b , is the ratio of retained earnings to earnings:

$$b = \frac{RE}{NI}$$

- Suppose a firm decided to distribute all its earnings in dividends (i.e., plowback ratio is $b = 0$), for the foreseeable future
- A firm uses the retained earnings to grow its assets and to generate more earnings in the future
- Without retained earnings, the firm cannot grow its assets, and thus its future earnings cannot grow
- If earnings don't grow, neither do dividends
- Thus, the dividend growth rate is $g = 0$, and the intrinsic value of the firm becomes:

$$\hat{P}_0 = \frac{EPS_1}{k},$$

where EPS_1 is the firm's *earnings per share* at period $t = 1$

- This ratio is known as the value of the firm's *assets in place*

- When the plowback ratio is strictly positive, $b > 0$, the firm retains earnings that it plans to use to grow its assets
- As a result, when $b > 0$, earnings grow and so do dividends
- In particular, when $b > 0$, the intrinsic value of the stock becomes:

$$\hat{P}_0 = \frac{EPS_1}{k} + PVGO,$$

where *PVGO* stands for *present value of growth opportunities*

- When should the firm retain earnings?
- It depends on the rate of return of the firm's growth opportunities, which is usually captured by measures such as *return on invested capital* (ROIC) or *return on equity* (ROE)
- Return on equity, *ROE*, is defined as the ratio of earnings to book equity, *BE*:

$$ROE = \frac{\text{Earnings}}{BE}$$

- It can be shown that when the firm uses its retained earnings to grow its assets, its dividends grow at a rate of

$$g = ROE \times b$$

- Thus, the intrinsic value of the stock becomes:

$$\hat{P}_0 = \frac{\bar{d}_1}{k - g} = \frac{EPS_1 \times (1 - b)}{k - ROE \times b}$$

- In particular, we note that:

- When $ROE = k$, then $\hat{P}_0 = \frac{EPS_1 \times (1 - b)}{k \times (1 - b)} = \frac{EPS_1}{k}$
- When $ROE > k$, then $\hat{P}_0 > \frac{EPS_1 \times (1 - b)}{k \times (1 - b)} = \frac{EPS_1}{k}$
- When $ROE < k$, then $\hat{P}_0 < \frac{EPS_1 \times (1 - b)}{k \times (1 - b)} = \frac{EPS_1}{k}$

- We summarize the previous result as follows

Effect of ROE on Stock Price

If ROE is strictly larger than the required cost of capital, k , the investment opportunities inside the firm are more valuable to investors than the investment opportunities outside the firm. In this case, the intrinsic value of the stock when firm retains earnings ($b > 0$) *increases* relative to the case when the firm does not retain any earnings. In particular, the firm should retain as much earnings as possible.

If ROE is smaller than k , then the intrinsic value of the stock when firm retains earnings ($b > 0$) *decreases* relative to the case when the firm does not retain any earnings. In this case, the firm should distribute all its earnings in dividends ($b = 0$).

- The dividend growth rate $g = ROE \times b$ is called the firm's *sustainable growth rate*

- We can use the previous analysis to calculate the price-to-earnings ratio (P/E ratio):

$$\frac{\hat{P}_0}{EPS_1} = \frac{1 - b}{k - ROE \times b}$$

- Because the right-hand side of this equation goes up when firm's investment opportunities are valuable (i.e., $ROE > k$), the P/E ratio is often used to gauge a firm's growth opportunities
- Note that EPS_1 is the earnings per share that the firm would earn in the future, at $t = 1$, which is not known
- The industry often likes to calculate and report a firm's P/E ratio, but they all use *historical* EPS not *future* EPS
- For this reason, you have to be careful when using P/E ratio for comparing the performance of a firm against other firms

- The P/E ratio is a *valuation multiple* that investor often use to calculate the value of a firm's stock price based on the stock prices of comparable firms
- For example, suppose we want to find the intrinsic value of firm M that generates EPS of \$1.5
- Suppose further that firms A, B and C are direct competitors with firm M, and their current P/E ratios are respectively, $(P/E)_A = 21$, $(P/E)_B = 22$, and $(P/E)_C = 26$
- What should be the intrinsic stock price of firm M?
- We first calculate the average P/E ratio across the compaable firms A, B, and C:

$$(P/E)_{Ave} = \frac{1}{3} \{ (P/E)_A + (P/E)_B + (P/E)_C \} = 23$$

- Then the intrinsic price of firm M's stock can be calculated by setting its P/E ratio equal to the average P/E ratio across the comparable firms

$$(P/E)_M = (P/E)_{Ave}$$

$$\frac{P_M}{1.5} = 23$$

- Solving for P_M we obtain, $P_M = \$1.5 \times 23 = \34.5
- Thus, more generally, the intrinsic stock price can be calculated with the formula:

$$P_M = EPS_M \times (P/E)_{Ave}$$

- Here's another example that combines the P/E multiple analysis with the discounted dividend model
- A stock just paid a dividend of \$1.1 per share. The dividend is expected to grow by 10% per year for the next 5 years. In year 6, the firm's payout ratio is expected to be 50% and its P/E ratio is expected to approach the historical P/E ratio of the average competitor in the industry, which currently sits at 20.
- If the stock's required rate of return is 15%, what should be the intrinsic stock price of the firm?

- An alternative approach to the discounted dividend model or P/E analysis values the firm using the firm's *free cash flows*
- Recall from your corporate finance class that a firm's free cash flows (FFCF) are the cash flows available to the firm's claim holders (equity and debt holders) after capital expenses
- A firm's FFCFs can be calculated with the formula:

$$FCFF = EBIT \times (1 - t_C) + Depreciation - CapEX - \text{Increase in } NWC,$$

where

- $EBIT$ = annual earnings before interest and tax
- $Depreciation$ = annual depreciation of property, plant and equipment (PP&E)
- $CapEx$ = capital expenditure (i.e., the increase in PP&E)
- NWC = net working capital
- t_C is corporate tax rate for this firm

- Recall the a firm's NWC can be calculated with the formula:

$$NWC = \text{Accounts receivable} + \text{Inventory} - \text{Accounts Payable}$$

- To value the firm's equity using FCFFs, we first value the entire firm, and then we subtract the firm's debt
- The value of the firm or the *enterprise value* of the firm, V , can be calculated with the formula:

$$V_0 = \frac{FCFF_1}{(1 + r_{WACC})^1} + \dots + \frac{FCFF_T}{(1 + r_{WACC})^T} + \frac{V_T}{(1 + r_{WACC})^T}$$

where

- T = the forecasting horizon
- V_T = terminal value or the enterprise value at the horizon
- r_{WACC} = weighted average cost of capital for the firm

- The value of the firm's equity, E_0 , is now calculated by subtracting the firm's debt, D_0 , and cash, $Cash_0$, from the enterprise value, V_0 :

$$E_0 = V_0 - D_0 + Cash_0$$

- The intrinsic stock price of the firm can then be calculated with:

$$\hat{P}_0 = \frac{E_0}{\# \text{Shares outstanding}}$$

- We next discuss how to calculate the terminal value, V_T , and the discount rate, $WACC$

- The terminal value, V_T , can be calculated in two ways
- If we believe the firm's FCFFs would grow at a constant rate of g from T onwards, then the terminal value can be calculated using the growing perpetuity formula:

$$V_T = \frac{FCFF_T \times (1 + g)}{r_{WACC} - g}$$

- As an alternative, we can calculate the terminal value using a valuation multiple
- The most common valuation multiple for FCFF is the Enterprise value-to-EBITDA multiple, $EV/EBITDA$, where
 - $EV = \text{Market equity} + \text{Debt} - \text{Cash}$
 - $EBITDA = \text{earnings before interest, taxes, depreciation and amortization}$

- If we can identify several firms that we can compare against, and if we know the EV/EBITDA multiple of these comparable firms at the horizon, T , then the enterprise value becomes:

$$V_T = EBITDA_T \times (EV/EBITDA)_{Ave},$$

where $(EV/EBITDA)_{Ave}$ is the average across the comparable firm's EV/EBITDA multiples

- In particular note that we need to know the EBITDA of the firm that we are valuing at time T , $EBITDA_T$, to be able to back out the enterprise value at the horizon, V_T

- The discount rate needed to discount back the FCFFs is calculated as follows:

$$r_{WACC} = (1 - L) \times r_E + (1 - t_C) \times L \times r_D$$

- $L = D/(D + E)$ is the firm's *desired* leverage ratio
 - r_E is the equity cost of capital
 - r_D is the debt cost of capital
 - t_C is the corporate tax rate
- The equity cost of capital, r_E can be calculated with the CAPM:

$$r_E = r_F + \beta_E(r_M - r_F)$$

- The debt cost of capital, r_D , can also be calculated using the CAPM, if we know the debt beta, β_D
- As an alternative, r_D can also be approximated with the yield of the firm's long-term debt

- The equity beta, β_E , depends on the desired leverage and the beta of the firm's assets, known as the unlevered beta, β_U
- The relationship between β_E (also known as levered beta) and β_U , the unlevered beta can be described with the Hamada equation:

$$\beta_E = \beta_U \times \left(1 + (1 - t_C) \times \frac{D}{E} \right)$$

- The asset beta, β_U , can be calculated from close competitors that are publically traded and that operate in the same industry as our firm
- For example, suppose there are 3 firms A, B, and C that are close competitors to our firm
- Because these firms are publically traded, we can calculate their equity betas, β_E^i , from their stock returns, where i stands for A, B or C

- Given the equity betas of the comparable firm i , β_E^i , and its debt-to-equity ratios, D^i/E^i , we can calculate the beta of firm i 's assets, β_U^i using the Hamada formula:

$$\beta_U^i = \frac{\beta_E^i}{1 + (1 - t_C) \times \frac{D^i}{E^i}},$$

- We now take the average across the asset betas of the three comparable firms, β_U^A , β_U^B , and β_U^C to calculate the asset beta for the firm that we are trying to value:

$$\beta_U = \frac{1}{3} (\beta_U^A + \beta_U^B + \beta_U^C)$$

- We illustrate with an example: See P1 in A6

- The same valuation principles that we used to value stocks can be used to value bonds
- We will focus on US Treasury bonds, but the analysis can be carried over to corporate or sovereign bonds as well
- A US Treasury bond has a number of characteristics:
 - T = Maturity of the bond can be anywhere between 30 days to 30 years
 - F = Face value represents the amount borrowed
 - c = Coupon rate represents the interest rate of the loan
 - n = Frequency of coupon payments in a year is semi-annual
- For example, suppose we want to value a 2-year US Treasury bond with face value \$1000, annual coupon rate of 6% paid semi-annually
- Then: $T = 2$, $F = 1000$, $c = 6\%$, and $n = 2$

- The bond will make $T \times n = 4$ semi-annual payments
- The semi-annual coupon payment equals:

$$\frac{C}{n} \times F = 30$$

- All US Treasury bonds repay the face value at the maturity in addition to the coupon payment
- Thus, the bond's payment at the maturity is:

$$\frac{C}{n} \times F + F = 1030$$

- The bond's cash flows can be summarized in a table:

Payment date	0	0.5	1	1.5	2
Bond CFs		30	30	30	1030

- When the price of the bond is known, we can calculate a bond's **yield-to-maturity (YTM)**
- A bond's YTM is the discount rate that equates the price of the bond to the present value of the bond's promised cash flows
- For example, if the bond in our example above trades currently at 105% of the face value (i.e., \$1,050), then the bond's YTM, y , is the solution to the equation:

$$1050 = \frac{30}{(1 + y/2)^{2 \times 0.5}} + \frac{30}{(1 + y/2)^{2 \times 1}} + \frac{30}{(1 + y/2)^{2 \times 1.5}} + \frac{30 + 1000}{(1 + y/2)^{2 \times 2}}$$

which becomes:

$$1050 = \frac{30}{(1 + y/2)^1} + \frac{30}{(1 + y/2)^2} + \frac{30}{(1 + y/2)^3} + \frac{30 + 1000}{(1 + y/2)^4}$$

- We can solve for y numerically in Excel using the IRR function

$$y/2 = IRR(-1050, 30, 30, 30, 1030)$$

which results in an YTM, $y = 2 \times 1.697\% = 3.393\%$

- The YTM is particularly easy to calculate for bonds that make no coupon payment, also known as **zero-coupon bonds**
- A zero-coupon bond makes only one payment, namely the face value of \$1000 at the bond's maturity
- Suppose a zero-coupon bond with maturity of 3 years trades currently at 94.5% of face value (i.e., \$945)
- Then the YTM of the bond, y is the solution to the equation:

$$945 = \frac{1000}{(1 + y/2)^{2 \times 3}}$$

- This equation can be solved analytically:

$$(1 + y/2)^6 = \frac{1000}{945}$$

which implies:

$$1 + y/2 = \left(\frac{1000}{945} \right)^{1/6} = 1.009473 \Rightarrow y = 2 \times 0.9473\% = 1.8946\%$$

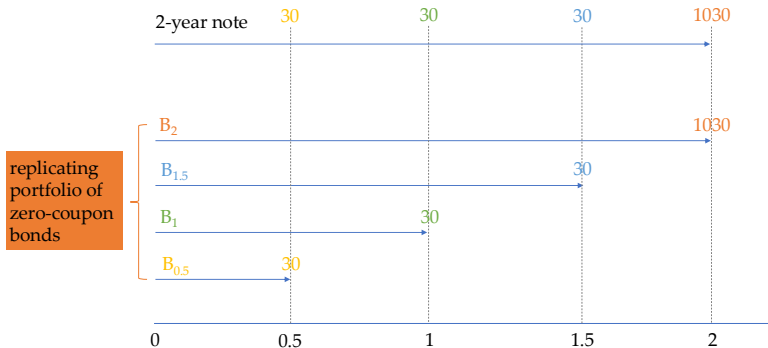
- Of course, we can also use the IRR function to calculate the YTM of this bond faster:

$$y/2 = IRR(-945, 0, 0, 0, 0, 0, 1000)$$

which results in $y = 2 \times 0.9473\% = 1.8946\%$.

- Zero-coupon bonds trade on the bond market under the acronym of **STRIPS**
- STRIPS are created by financial intermediaries by "stripping out" each coupon payment and the principal payment from a US Treasury note or bond and selling them to investors

- There is a close relationship between zero-coupon bonds and a coupon-paying US Treasury bond
- A coupon-paying US Treasury bond can be viewed as a portfolio of zero-coupon bonds
- To illustrate, recall that a 2-year US Treasury note with a 6% coupon makes 4 semi-annual cash flow payments of 30, 30, 30, and 1030, respectively
- We can build a portfolio of 4 zero-coupon bonds that makes exactly the same cash flow payments as the 2-year bond above:
 - Buy long 30/1000 of a 6-month zero-coupon bond, say $B_{0.5}$
 - Buy long 30/1000 of a 1-year zero-coupon bond, say B_1
 - Buy long 30/1000 of a 1.5-year zero-coupon bond, say $B_{1.5}$
 - Buy long 1030/1000 of a 2-year zero-coupon bond, say B_2



- Note that the portfolio of zero-coupon bonds generates the following cash flows:
 - In 0.5 years, $B_{0.5}$ position pays off $0.03 \times 1000 = 30$
 - In 1 years, B_1 position pays off $0.03 \times 1000 = 30$
 - In 1.5 years, $B_{1.5}$ position pays off $0.03 \times 1000 = 30$
 - In 2 years, B_2 position pays off $1.03 \times 1000 = 1030$
- These are exactly the cash flows of our 2-year US Treasury note
- Because the zero-coupon bond portfolio matches perfectly the cash flows of the 2-year note, we call it a **replicating portfolio**
- In finance, the **law of one price (LOOP)** says that two assets that have the same exact cash flows must have the same price
- In particular, the price of the 2-year note must equal the cost of the replicating portfolio

- The cost of the replicating portfolio is the money we have to pay to set up the 4 positions
- Suppose the 4 zero-coupon bonds trade currently at the following prices:
 - $B_{0.5}$ trades at 99.6%
 - B_1 trades at 97.8%
 - $B_{1.5}$ trades at 95.9%
 - B_2 trades at 93.4%
- Then the cost of the replicating portfolio becomes:

$$\text{Cost} = 0.03 \times 996 + 0.03 \times 978 + 0.03 \times 959 + 1.03 \times 934 = \$1050$$

- Thus, according to LOOP, the 2-year note should be trading at exactly \$1050

- What would happen if the 2-year note were to trade for a different price?
- If that were the case, we would have an **arbitrage opportunity** in the bond market: i.e., *the opportunity to make risk-free profits*
- For example, suppose the 2-year bond was trading at \$1060
- A smart investor could make a quick risk-free profit with a long-short strategy:
 - **Short-sell** the 2-year bond to get \$1060 right now
 - **Buy long** the replicating portfolio of the 4 zero-coupon bonds, which should cost you \$1050 right now
- Recall that when you short-sell a bond, you have to return to the broker all the coupon payments that the bond pays out
- Where would the investor get the money to repay those coupons?

- The money would come from investor's long position, namely the 4 zero-coupon bonds
- The 2-year note's coupon payment of \$30 in 6 months:
 - Will be perfectly offset by investor's $B_{0.5}$ position in the replicating portfolio, which pays off \$30 exactly in 6 months
- The 2-year note's coupon payment of \$30 in 1 year:
 - Will be perfectly offset by investor's B_1 position in the replicating portfolio, which pays off \$30 exactly in 1 year
- The 2-year note's coupon payment of \$30 in 1.5 years:
 - Will be perfectly offset by investor's $B_{1.5}$ position in the replicating portfolio, which pays off \$30 exactly in 1.5 years
- The 2-year note's final payment of \$1030 in 2 years:
 - Will be perfectly offset by investor's B_2 position in the replicating portfolio, which pays off \$1030 exactly in 2 years

- Thus, the investor's replicating portfolio will fully repay the 2-year note borrowed from the broker
- In particular, the investor owes nothing to the broker
- Meanwhile, the investor just made $\$10 = \$1060 - \$1050$ in his account
- This *profit* is completely *risk-free*
- This is how arbitrage opportunities are exploited: using long-short strategies
- Of course, there are many smart investor who may have spotted the same arbitrage opportunity
- As more and more investors try to take advantage of the arbitrage opportunity, prices of the 2-year bond and the zero-coupon bonds adjust to the point where the arbitrage opportunity disappears

- The example above illustrates how to set the price on a 2-year note when we know the prices of zero-coupon bonds
- This result is more general: if the prices of zero-coupon bonds are known, we can price any US Treasury coupon-paying bond
- The YTM of the zero-coupon bonds can be calculated fairly quickly using the IRR function:
 - For $B_{0.5}$, YTM is $y_{0.5} = 2 \times \text{IRR}(-996, 1000) = 0.803\%$
 - For B_1 , YTM is $y_1 = 2 \times \text{IRR}(-978, 0, 1000) = 2.237\%$
 - For $B_{1.5}$, YTM is $y_{1.5} = 2 \times \text{IRR}(-959, 0, 0, 1000) = 2.811\%$
 - For B_2 , YTM is $y_2 = 2 \times \text{IRR}(-934, 0, 0, 0, 1000) = 3.443\%$
- The plot of the yields $y_{0.5}$, y_1 , $y_{1.5}$, and y_2 is known as the **US Treasury spot rate curve**
- It is usually provided in a table:

Maturity	0.5	1	1.5	2
Spot rates	0.803%	2.237%	2.811%	3.443%

- When the US Treasury spot rate curve is known, we can price any US Treasury coupon paying bond
- For example, suppose the US Treasury spot rate curve is the one above
- What should be the price of a US Treasury bond with 7% coupon (paid semi-annually) and maturity of 18 months (i.e., 1.5 years)?
- Bond cash flows and discounted cash flows (DCF) become:

Maturity	0.5	1	1.5	2
Spot rates	0.803%	2.237%	2.811%	3.443%
Bond CFs	35	35	1035	0
Bond DCFs	$\frac{35}{(1 + \frac{0.803\%}{2})^{2 \times 0.5}}$ = 34.86	$\frac{35}{(1 + \frac{2.237\%}{2})^{2 \times 1}}$ = 34.23	$\frac{1035}{(1 + \frac{2.811\%}{2})^{2 \times 1.5}}$ = 992.565	0

- Price of the bond becomes: $34.86 + 34.23 + 992.565 = 1061.655$

- Where do the spot rates or the prices of zero-coupon bonds come from?
- They could come from STRIPS that are currently trading
- They could also come from coupon-paying US Treasury bonds that are currently trading
- Effectively, we do our own stripping through a process called *bootstrapping*
- We illustrate how bootstrapping works with an example
- Suppose we observe the following bond prices:
 - 1-year bond with 5% coupon trades at 97.7%
 - 1-year bond with 6% coupon trades at 98.64%
 - 2-year bond with 7% coupon trades at 94.425%
 - 2-year bond with 8% coupon trades at 96.2%
- What should be the prices of the zero-coupon bonds $B_{0.5}$, B_1 , $B_{1.5}$, and B_2 that are consistent with the prices above?

- Recall from our analysis above that each coupon-paying bond can be replicated with a portfolio of the 4 zero-coupon bonds
- Recall also that according to LOOP the price of the coupon-paying bond must equal the cost of the corresponding replicating portfolio
- We then have:

$$977 = 2.5\%B_{0.5} + 102.5\%B_1 + 0\%B_{1.5} + 0\%B_2$$

$$986.4 = 3\%B_{0.5} + 103\%B_1 + 0\%B_{1.5} + 0\%B_2$$

$$944.25 = 3.5\%B_{0.5} + 3.5\%B_1 + 3.5\%B_{1.5} + 103.5\%B_2$$

$$962 = 4\%B_{0.5} + 4\%B_1 + 4\%B_{1.5} + 104\%B_2$$

- Need to solve for the unknown prices $B_{0.5}$, B_1 , $B_{1.5}$, and B_2

- The 4 equations above can be re-written in matrix form:

$$\begin{pmatrix} 977 \\ 986.4 \\ 944.25 \\ 962 \end{pmatrix} = \begin{pmatrix} 2.5\% & 102.5\% & 0\% & 0\% \\ 3\% & 103\% & 0\% & 0\% \\ 3.5\% & 3.5\% & 3.5\% & 103.5\% \\ 4\% & 4\% & 4\% & 104\% \end{pmatrix} \times \begin{pmatrix} B_{0.5} \\ B_1 \\ B_{1.5} \\ B_2 \end{pmatrix}$$

- The solution can be found by *inverting* (i.e., "dividing") the matrix (just like in a linear equation):

$$\begin{pmatrix} B_{0.5} \\ B_1 \\ B_{1.5} \\ B_2 \end{pmatrix} = \begin{pmatrix} 2.5\% & 102.5\% & 0\% & 0\% \\ 3\% & 103\% & 0\% & 0\% \\ 3.5\% & 3.5\% & 3.5\% & 103.5\% \\ 4\% & 4\% & 4\% & 104\% \end{pmatrix}^{-1} \times \begin{pmatrix} 977 \\ 986.4 \\ 944.25 \\ 962 \end{pmatrix}$$

- The inverse of a matrix can be calculated in Excel with the function MINVERS()

- We obtain:

$$\begin{pmatrix} B_{0.5} \\ B_1 \\ B_{1.5} \\ B_2 \end{pmatrix} = \begin{pmatrix} -206 & 205 & 0 & 0 \\ 6 & -5 & 0 & 0 \\ 200 & -200 & -208 & 207 \\ 0 & 0 & 8 & -7 \end{pmatrix} \times \begin{pmatrix} 977 \\ 986.4 \\ 944.25 \\ 962 \end{pmatrix}$$

- Using the MMULT() function in Excel to calculate the right-hand side of the equation above, we obtain:

$$\begin{pmatrix} B_{0.5} \\ B_1 \\ B_{1.5} \\ B_2 \end{pmatrix} = \begin{pmatrix} 950 \\ 930 \\ 850 \\ 820 \end{pmatrix}$$

- That is $B_{0.5} = 950$, $B_1 = 930$, $B_{1.5} = 850$, and $B_2 = 820$

- The YTM of the zero-coupon bonds can now be calculated using the IRR function:
 - For $B_{0.5}$, YTM is $y_{0.5} = 2 \times \text{IRR}(-950, 1000) = 10.53\%$
 - For B_1 , YTM is $y_1 = 2 \times \text{IRR}(-930, 0, 1000) = 7.39\%$
 - For $B_{1.5}$, YTM is $y_{1.5} = 2 \times \text{IRR}(-850, 0, 0, 1000) = 11.11\%$
 - For B_2 , YTM is $y_2 = 2 \times \text{IRR}(-820, 0, 0, 0, 1000) = 10.17\%$
- The US Treasury spot rate curve becomes

Maturity	0.5	1	1.5	2
Spot rates	10.53%	7.39%	11.11%	10.17%

- We can use this spot rate curve to calculate the price of any US Treasury coupon-paying bond with a maturity of at most 2 years